Score is Not Circuit: Evaluating Mechanistic Localization as a Score-to-Circuit Stack

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Abstract

Mechanistic localization methods are often assessed through component-level importance estimates, yet downstream claims rely on circuits induced from those scores under particular selection rules, evaluation metrics, and compute budgets. We argue that score quality and circuit quality are distinct objects, and we formalize mechanistic localization as a four-layer score-to-circuit stack consisting of scoring, calibration, selection, and evaluation. We instantiate this view on edge-level circuit localization and study how scorer corrections, uncertainty-aware calibration, and metric-aware selectors interact across models and tasks in MIB. Our analysis reveals three recurrent gaps: an estimation gap between approximate and exact edge scores, an interaction gap between edgewise importance and multi-edge circuit faithfulness, and an objective gap between circuits favored by performance-oriented and model-matching metrics. Across this stack, local scorer improvements do not compose monotonically into better circuits: a scorer can become more faithful to exact edge effects while yielding worse downstream circuits, and global uncertainty shrinkage can severely degrade circuit quality. These findings support a simple conclusion: mechanistic localization should be evaluated as a full score-to-circuit pipeline rather than as an isolated scoring problem. We conclude by proposing compute-aware, metric-aware reporting standards for future work on circuit discovery and localization.

1 Introduction

Mechanistic localization assigns importance scores to internal components, while downstream claims usually concern *circuits*: budgeted subgraphs produced by ranking, calibration, thresholding, or reranking those scores under explicit evaluation metrics and compute constraints [W, C, S, K]. The gap between score fields and reported circuits now shapes empirical comparisons. Exact and approximate patching methods have improved [K, H, J], MIB moved evaluation from isolated case studies to explicit circuit metrics on fixed model–task cells [M], and the BlackboxNLP shared task showed that selection and regularization choices can materially change the circuits induced from the same score field [A].

These developments place the full score-to-circuit pipeline at the level of the scientific unit: the object that turns localization scores into evaluated circuit claims.

This paper develops an evaluation framework for mechanistic localization claims as *score-to-circuit pipelines.* We model each pipeline as a four-layer stack:

scorer $\rightarrow$ **calibrator** $\rightarrow$ **selector** $\rightarrow$ **evaluator.**

The guiding claim is:

A score field becomes a circuit claim only through the full stack.

Circuit quality is therefore a property of the scorer, calibrator, selector, and evaluator together.

We instantiate the score-to-circuit pipeline on edge-level circuit localization in MIB [M] by comparing exact edge patching, edge attribution patching, integrated-gradient scoring, and three edge-level nonlinear corrections inspired by AtP* [K]. The evaluation then compares annotation-only calibration with rewrite calibration and applies metric-aware selectors, including stable shortlists, shortlist-local lower-confidence reranking, and true PNR. Across the evaluated cells, agreement with exact edge effects has limited predictive power for downstream circuit quality, local scorer repairs compose nonmonotonically, global uncertainty rewrites can collapse strong raw rankings, and shortlist-based selectors often recover them by using uncertainty locally. Supported CPR winners and model-matching winners diverge repeatedly.

The resulting protocol formalizes mechanistic localization as a score-to-circuit stack, separates recurrent gaps in localization claims, and treats compute as a first-class reporting axis. It clarifies the metric geometry of circuit evaluation by distinguishing positive-performance quality, magnitude-mode area, and circuit–model distance. The accompanying MIB study shows that scorer quality, selector quality, and circuit quality are materially different objects. The released artifact makes the protocol auditable through proofs, executable witnesses, and mechanized checks.

Section 2 defines the stack, the metric geometry, and the structural results. Section 3 specifies the scorers, calibrators, selectors, and evaluation stages used in the empirical study. Section 4 presents the main findings. Section 5 distills the implications for future evaluation practice.

2 The Score-to-Circuit Stack

Definition 1 (Localization stack)**.** *A localization stack is a quadruple* $\mathfrak{L} = (S, C, A, \mathrm{Eval})$ *consisting of a scorer* S*, a calibrator* C*, a selector* A*, and an evaluator* Eval*.*

Given a model M, a clean/counterfactual dataset D, and an edge graph $G = (V, \mathcal{E})$, the scorer returns raw signed scores

$$S(M, D, G) = s \in \mathbb{R}^{|\mathcal{E}|}.$$

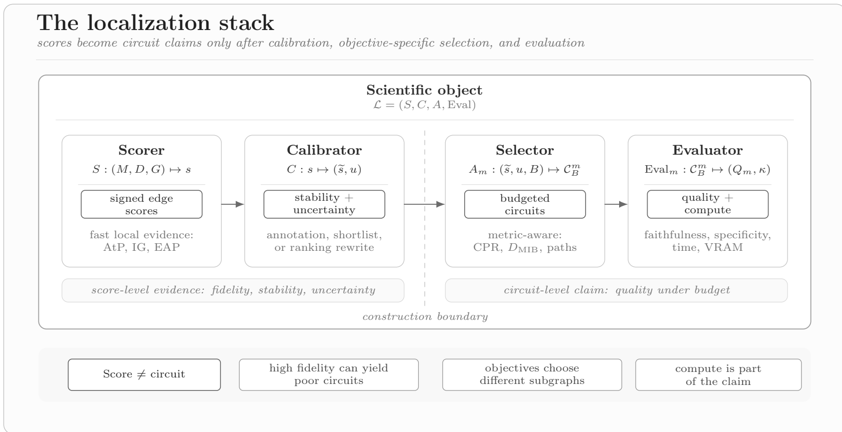

Figure 1: The localization stack. Mechanistic localization claims are produced by a pipeline that maps raw edge scores to budgeted circuits and then evaluates those circuits under objective-specific metrics and compute constraints.

The calibrator maps raw scores to

$$C(s) = (\tilde{s}, \rho, \sigma),$$

where $\tilde{s}$ is a possibly rewritten score vector, ρ records stability information, and σ records uncertainty information. For an objective m, the selector constructs a family of budgeted circuits

$$A_m(\tilde{s}, \rho, \sigma) = \{C_k^m \subseteq \mathcal{E} : k \in K\},$$

where K is the benchmark budget grid. Finally, the evaluator returns circuit quality and cost,

$$\mathrm{Eval}_m(\{C_k^m\}) = Q_m, \qquad \mathrm{Cost}(\mathfrak{L}) = (\text{wall time}, \text{VRAM}, \#F, \#B).$$

2.1 Metric geometry

Positive-performance selection uses

$$u_e^+(s) = \max(s_e, 0),$$

and induces a faithfulness curve $f^+(k)$ whose integrated area is the circuit performance ratio

$$\mathrm{CPR} = \int f^+(k)\, d\mu(k),$$

where μ is the normalized discrete measure on the benchmark budget grid. Magnitude-mode selection uses

$$u_e^{\mathrm{mag}}(s) = |s_e|,$$

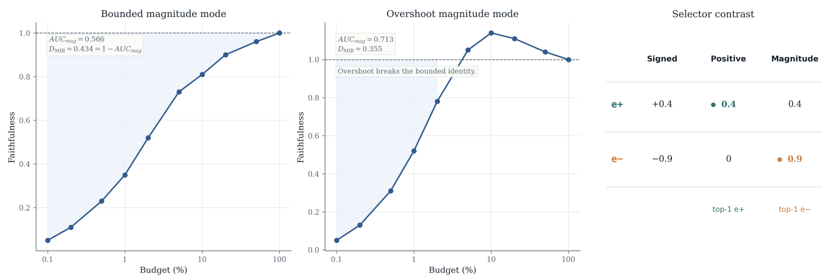

Figure 2: Metric geometry for circuit evaluation. Circuit evaluation separates performance-oriented quality (CPR), the magnitude-mode area ($\mathrm{AUC}_{\mathrm{mag}}$), the benchmark distance ($\mathrm{D}_{\mathrm{MIB}}$), and the derived model-matching quality ($\mathrm{MMQ} = 1 - \mathrm{D}_{\mathrm{MIB}}$). In the no-overshoot regime $f^{\mathrm{mag}}(k) \leq 1$, $\mathrm{D}_{\mathrm{MIB}} = 1 - \mathrm{AUC}_{\mathrm{mag}}$ and therefore $\mathrm{MMQ} = \mathrm{AUC}_{\mathrm{mag}}$.

and induces a curve $f^{\mathrm{mag}}(k)$ with auxiliary area

$$\mathrm{AUC}_{\mathrm{mag}} = \int f^{\mathrm{mag}}(k)\, d\mu(k).$$

The benchmark distance is

$$\mathrm{D}_{\mathrm{MIB}} = \int |1 - f^{\mathrm{mag}}(k)|\, d\mu(k),$$

with lower values preferred. For readability in winner tables and transfer tables, we also report the derived high-is-good quantity

$$\mathrm{MMQ} = 1 - \mathrm{D}_{\mathrm{MIB}}.$$

Proposition 1 (Distance–area relation)**.** *Assume that μ is normalized. Then*

$$\mathrm{D}_{\mathrm{MIB}} = 1 - \mathrm{AUC}_{\mathrm{mag}} + 2 \int \left(f^{\mathrm{mag}}(k) - 1\right)_{+} d\mu(k).$$

Hence, if $f^{\mathrm{mag}}(k) \leq 1$ for all budgets, then $\mathrm{D}_{\mathrm{MIB}} = 1 - \mathrm{AUC}_{\mathrm{mag}}$ and therefore $\mathrm{MMQ} = \mathrm{AUC}_{\mathrm{mag}}$. The correction term is exactly twice the total overshoot above 1.

CPR and MMQ define different geometric targets. Even within a fixed scorer family, their optima can occur at different stacks.

2.2 Four gaps

The stack induces four analytically distinct gaps. The *estimation gap* measures disagreement between approximate and exact edge scores on tractable cells. The *rank-distortion gap* measures how much a calibrator changes the raw ordering. The *interaction gap* measures the discrepancy between additive edgewise utility and realized multi-edge circuit faithfulness. The *objective gap* measures the disagreement between stacks favored by CPR and by model-matching quality. Compute is tracked throughout as a reporting axis because stack quality is meaningful only for materialized cells.

Two structural results organize the rest of the paper.

Proposition 2 (Invariance under isotonic calibration). *Let A_k be a deterministic top-k selector with fixed tie-breaking, and let ϕ be strictly increasing. Then*

$$A_k(\{\phi(u_e)\}) = A_k(\{u_e\})$$

for every budget k.

Rank-preserving calibration is separated from calibration procedures that rewrite the selection order. Annotation calibration preserves the raw ranking while allowing metadata-aware selectors to use ρ or σ near the selection boundary. Rewrite calibration can change the selected circuit through the induced utility order, tie structure, or selector inputs.

Proposition 3 (Zero interaction gap under exact additivity). *Assume there exists a non-negative edge utility system w^m such that every circuit's faithfulness equals its normalized additive mass under w^m. Then the additive top-k proxy matches the realized faithfulness curve exactly, the interaction gap is zero, and the curve is monotone non-decreasing.*

Observed interaction gaps indicate limits of the additive score-to-circuit approximation. In practice, they may combine true multi-edge interaction, scorer estimation error, and selector–objective mismatch. The relevant evaluation unit is therefore the full stack.

3 Protocol and Experimental Setting

We instantiate the stack on edge-level circuit localization in MIB [M]. The supported matrix contains seven circuit-localization cells with both CPR and model-matching evaluation, together with InterpBench for edge-level AUROC [G].

3.1 Concrete stack instantiations

The scorer family consists of EAP, EAP-IG, exact edge patching on tractable cells, and three local nonlinear corrections inspired by AtP* [K]. In concise form,

$$s_e^{\text{EAP-QK}} = \langle \Delta a_e, \nabla_a L \rangle, \qquad s_e^{\text{EAP-MLP}} = \langle \Delta \text{MLP}_e, \nabla_{\text{MLP}} L \rangle,$$

and EAP* applies both local corrections where the architecture supports them. Calibration is split into two classes:

- **annotation-only:** $\tilde{s}_e = s_e$, with (ρ_e, σ_e) exposed to the selector;

Item	What is reported
Cells	model, task, split/stage
Scorers	EAP, EAP-IG, Exact, EAP-QK, EAP-MLP, EAP*
Calibration	annotation-only vs. rewrite
Selectors	raw top-k, stable shortlist, LCB shortlist, true PNR
Hyperparameters	shortlist factor α, threshold τ, penalty λ, mixture p
Metrics	CPR $\uparrow$, $\mathrm{AUC}_{\mathrm{mag}}$ $\uparrow$, $\mathrm{D}_{\mathrm{MIB}}$ $\downarrow$, MMQ $\uparrow$, AUROC$\uparrow$
Compute	wall time, VRAM, bounded-probe status

Table 1: Reporting schema for score-to-circuit evaluation. Rows are stratified by scorer-level runs, selector development runs, frozen validation, and public-test replay.

- **rewrite:** the score field is modified, for example

$$\tilde{s}_e = \mathrm{sign}(s_e)\max(|s_e|\rho_e - \lambda\sigma_e, 0).$$

Selectors include signed top-k, magnitude top-k, true PNR, stable shortlists, and shortlist-local lower-confidence reranking. Stable shortlist methods protect the raw top-αk region and apply stability or uncertainty information inside that shortlist. LCB-within-shortlist uses

$$r_e^m = u_e^m - \lambda\sigma_e$$

within the protected candidate set.

Compute accounting. Compute is part of the evaluation target. A method–cell pair is classified as *materialized* when it yields score files and downstream circuit evaluation under the declared budget. The remaining attempted pairs are classified as bounded probes or unsupported cells. Reported wall-clock times are end-to-end stack times for the corresponding metric evaluation.

Exact cells and frozen transfer. Exact edge patching is restricted to tractable and scientifically interpretable cells, namely GPT-2+IOI and InterpBench+IOI. Selector hyperparameters are chosen on a bounded development grid. A small number of stacks are then frozen and replayed on held-out validation and public test.

3.2 Selected comparisons

Cell	Stage	CPR ↑ winner	MMQ ↑ winner
GPT-2 + IOI	supported val.	EAP-IG	EAP*
Gemma-2 + MCQA	supported val.	EAP-MLP	EAP-MLP
Gemma-2 + ARC-E	supported val.	EAP-MLP	EAP
Llama-3.1 + MCQA	supported val.	EAP-MLP	EAP*
Llama-3.1 + ARC-E	supported val.	EAP-IG	EAP
Qwen2.5 + IOI	supported val.	EAP-IG	EAP*
Qwen2.5 + MCQA	supported val.	EAP-IG	EAP-IG
InterpBench + IOI	supported val.	AUROC winner:	EAP-IG

Table 2: Metric winners on the supported validation matrix under the default scorer-level selector regime. The top stack for CPR and the top stack for MMQ differ in five of the seven circuit cells, showing that method rankings are objective-dependent.

Cell	Frozen stack	Objective	Split	Quality	Time (s)
GPT-2 + IOI	EAP-MLP + stable shortlist	CPR ↑	val.	1.339	680
GPT-2 + IOI	same	CPR ↑	public	1.283	—
Qwen2.5 + IOI	EAP* + LCB shortlist	CPR ↑	val.	1.607	1589
Qwen2.5 + IOI	same	CPR ↑	public	1.604	—
Qwen2.5 + IOI	EAP* + LCB shortlist	MMQ ↑	val.	0.968	1579
Qwen2.5 + IOI	same	MMQ ↑	public	0.969	—
Qwen2.5 + MCQA	EAP* + stable shortlist	MMQ ↑	val.	0.922	193
Qwen2.5 + MCQA	same	MMQ ↑	public	0.919	—
Gemma-2 + MCQA	EAP-MLP + true PNR	CPR ↑	val.	1.696	274
Gemma-2 + MCQA	same	CPR ↑	public	1.720	—

Table 3: Selected frozen validation rows and public-test replay. Rows are reported when split, selector, and objective are unambiguous.

4 Results

4.1 Exact agreement and circuit quality diverge

We first examine the exact-tractable cells. On GPT-2+IOI, EAP-IG improves signed Pearson agreement with exact edge effects relative to EAP (0.931 vs. 0.814), while the strongest scorer-level CPR in that family comes from EAP-MLP rather than EAP-IG (2.177 vs. 2.087). On InterpBench, EAP-IG delivers the best AUROC, while EAP* collapses despite combining two local corrections. Exact one-edge fidelity and induced circuit quality therefore produce different empirical rankings.

The scorer ablation also clarifies the role of local nonlinear corrections. EAP-QK functions as a negative control: it provides little downstream rescue and degrades sharply on InterpBench. EAP-MLP is the strongest scorer-level rescue in this campaign, improving CPR over EAP on GPT-2+IOI and Qwen2.5+MCQA while remaining competitive on model matching. EAP* is useful but mixed: the combined scorer composes its local corrections nonmonotonically.

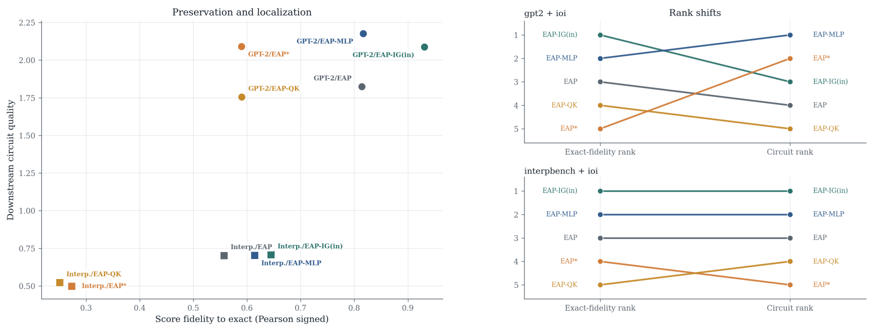

Figure 3: Exact-score fidelity versus downstream circuit quality on tractable cells. Agreement with exact edge effects and induced circuit quality yield different empirical rank orders.

4.2 Interaction and objective gaps are structural

To evaluate the interaction gap, we compare complete stacks rather than isolated edge scores. Under exact additive composition, top-k utility mass would predict the realized faithfulness curve and the gap would vanish by Proposition 3. The empirical curves retain substantial residual gaps, indicating that a simple additive score-to-circuit approximation is insufficient. In practice, the residual may combine true multi-edge interaction, scorer estimation error, and selector–objective mismatch, so edgewise score quality alone leaves out information needed to predict circuit quality.

For CPR and MMQ, Table 2 shows different top stacks in five of the seven supported circuit cells. The disagreement is especially sharp on GPT-2+IOI and Qwen2.5+IOI, where EAP-IG is strongest for CPR but EAP* yields the strongest model-matching circuits. Separating performance-oriented and model-matching objectives changes the selected stack.

4.3 Global uncertainty rewrites can collapse strong scorers

Annotation and rewrite calibration induce different stack behavior. Annotation-only calibration preserves the raw score order and exposes metadata for boundary decisions. Rewrite calibration globally modifies the score field. The most dramatic case is Qwen2.5+IOI with EAP*. The raw-selector stack achieves CPR $= 1.607$ and MMQ $= 0.968$ on frozen validation and transfers almost unchanged to public test (1.604, 0.969). The corresponding rewrite collapses to CPR ≈ 0.25 and MMQ ≈ 0.25 on validation. Same-slice controls isolate the rewrite stage as the driver of the collapse.

This failure motivates a safer selector design: use uncertainty inside a protected candidate set. Stable shortlists and shortlist-local lower-confidence reranking recover several of the strongest cells because they preserve the raw support

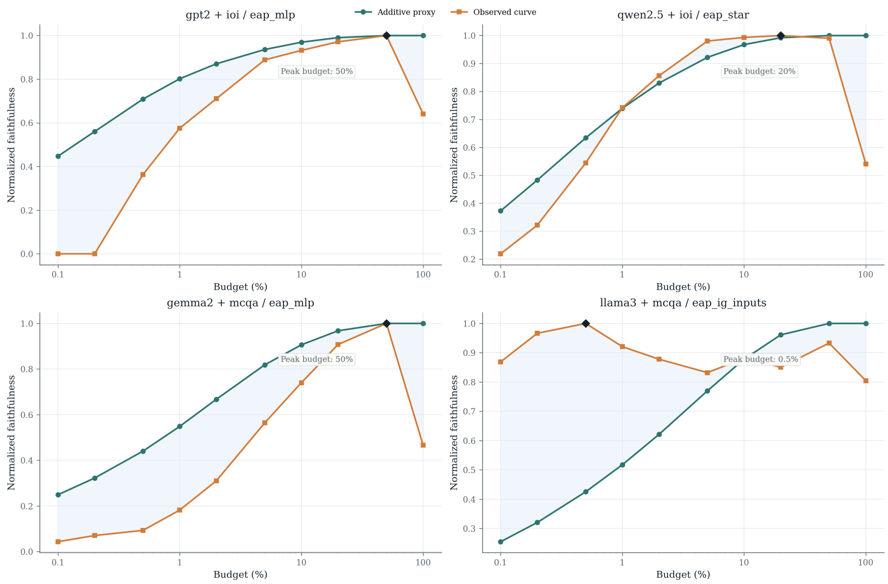

Figure 4: Representative interaction-gap cases. Each panel compares an additive edgewise proxy with the realized circuit faithfulness curve. The gaps are large, non-monotone, and budget-dependent, showing where additive score-to-circuit translation breaks down.

and apply uncertainty near the selection boundary. In several top rows, this selector-level decision changes circuit quality more than marginal scorer improvements.

4.4 Compute changes the scientific recommendation

Stack comparisons depend on the compute regime. On the supported matrix, exact patching remains indispensable on tractable cells, while runtime and memory restrict its scope. EAP-IG often provides the strongest raw CPR, but selector design and frozen replay can change its practical advantage. Concrete examples illustrate the tradeoff. On InterpBench+IOI, EAP-IG improves AUROC over EAP while also being faster on the supported-matrix rerun (0.747 in 48.9s versus 0.726 in 56.9s). On GPT-2+IOI, EAP-MLP reaches a stronger scorer-level CPR than EAP-IG at comparable end-to-end cost (2.177 in 1313s versus 2.087 in 1505s). On Qwen2.5+MCQA, by contrast, EAP-IG retains an advantage in both CPR and MMQ while being slightly faster than the nonlinear variants.

Across the evaluated cells, score-to-circuit behavior depends on the full resource-bounded stack. Raw scorers paired with robust selectors are strongest in some settings, different scorers are favored by different objectives in others, and better agreement with exact edge effects can still yield weaker circuits. These patterns are stack-level properties and are not captured by scorer-only

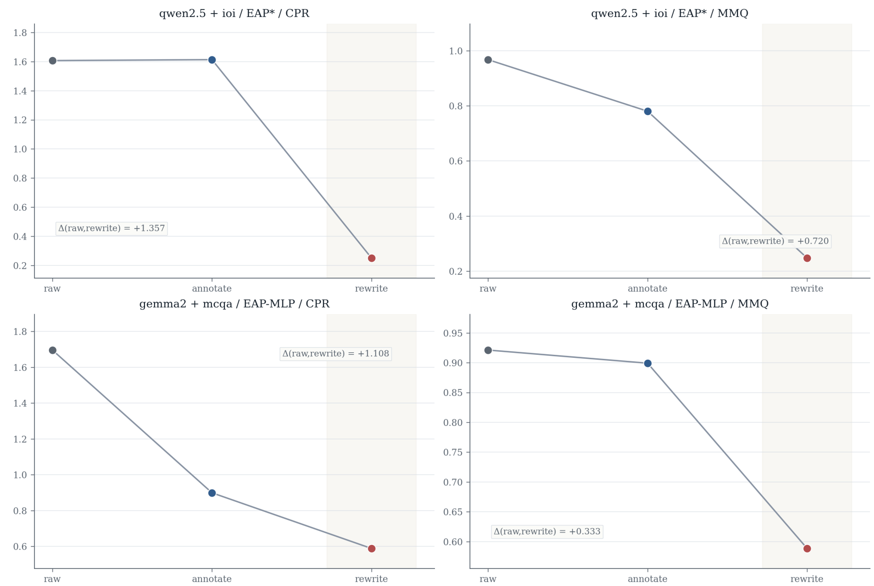

Figure 5: Calibration collapse and local rescue. Global rewrite can collapse strong raw scorers, whereas selectors that spend uncertainty within a protected shortlist preserve or recover several of the strongest cases. The sharpest collapse appears for Qwen2.5+IOI with EAP*.

summaries.

5 Discussion: From Scorers to Stack Evaluation

The empirical study supports a change in reporting practice for mechanistic localization.

Stack-level claims. Local scorer improvements can compose nonmonotonically into downstream circuits. This is clearest for EAP-QK and for the mixed behavior of EAP*. Once circuits are selected, evaluated under metric-specific objectives, and constrained by compute, scorer-level summaries provide only one layer of the localization claim.

Exact edge fidelity as a causal reference. Exact patching provides the right one-edge causal reference on tractable cells. Circuit quality also depends on objective choice, selector design, and multi-edge composition. As a result, stronger agreement with exact edge effects can coexist with weaker induced circuits.

Calibration type. Annotation and rewrite are different scientific operations. Annotation-only calibration preserves the raw score field and exposes metadata for selection. Rewrite calibration globally changes the candidate order and can remove the useful support of a strong raw ranking. Localization papers should state whether uncertainty is used as annotation, as a local shortlist reranker, or as a global rewrite.

Metric orientation. This paper distinguishes CPR $\uparrow$, $\mathrm{AUC}_{\mathrm{mag}}$ $\uparrow$, $\mathrm{D}_{\mathrm{MIB}}$ $\downarrow$, and MMQ $\uparrow$. These quantities answer different geometric questions. In particular, model-matching quality and raw magnitude-mode area diverge once overshoot is admitted. Reporting them separately makes the objective gap visible.

Compute accounting. Claims about localization quality depend on what was materialized. Reporting should include the scorer, calibrator type, selector, objective, split/stage, wall time, and any bounded-probe or unsupported status. These fields identify the scientific object being evaluated.

These findings suggest the following reporting discipline:

1. Report the full stack (S, C, A, Eval) together with the scorer.
2. Distinguish annotation-only calibration from rewrite calibration.
3. Separate CPR, $\mathrm{AUC}_{\mathrm{mag}}$, $\mathrm{D}_{\mathrm{MIB}}$, and MMQ explicitly.
4. Use exact edge patching as a causal reference on tractable cells.
5. Report compute and admissibility together with quality.

The companion artifact supports this discipline as an auditability device. It includes full prose proofs, executable theorem witnesses, and mechanized checks for the central structural invariants.

6 Conclusion

Mechanistic localization claims reach empirical use as circuit claims: budgeted subgraphs selected from score fields under metric and compute constraints. We formalize this evaluation unit as a score-to-circuit stack, separating scorer, calibrator, selector, and evaluator, and distinguishing performance-oriented quality from model-matching quality. Under this formulation, localization claims can change through score estimation, rank transformation, multi-edge composition, objective choice, and materialized compute.

In the MIB-grounded study, these layers change the scientific conclusion. Exact-score agreement and downstream circuit quality rank methods differently; CPR and MMQ winners diverge in five of the seven supported circuit cells; local nonlinear corrections compose nonmonotonically; and global uncertainty rewrite can collapse a strong raw stack, while shortlist-based selectors can preserve and transfer useful support. Compute further determines which stacks are admissible for a given model–task cell.

Mechanistic localization papers should therefore report the full stack (S, C, A, Eval), together with the calibration type, selector, objective, split or stage, quality metric, and compute status. This reporting unit makes localization claims comparable, auditable, and faithful to the object ultimately being evaluated: circuits induced from scores under declared constraints.

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A Proofs of the structural results

This appendix records the mathematical proofs used in the main text. The companion artifact additionally contains executable witnesses and mechanized checks.

A.1 Distance–area relation

Assume that μ is normalized. Write $f(k) = f^{\mathrm{mag}}(k)$ and decompose

$$|1 - f(k)| = \begin{cases} 1 - f(k), & f(k) \le 1, \\ f(k) - 1, & f(k) > 1. \end{cases}$$

Equivalently,

$$|1 - f(k)| = 1 - f(k) + 2(f(k) - 1)_+.$$

Integrating both sides against μ gives

$$\mathrm{D_{MIB}} = \int |1 - f(k)| \, d\mu(k) = 1 - \int f(k) \, d\mu(k) + 2 \int (f(k) - 1)_+ \, d\mu(k).$$

Since $\int f(k) \, d\mu(k) = \mathrm{AUC_{mag}}$, we obtain

$$\mathrm{D_{MIB}} = 1 - \mathrm{AUC_{mag}} + 2 \int (f^{\mathrm{mag}}(k) - 1)_+ \, d\mu(k).$$

If $f^{\mathrm{mag}}(k) \leq 1$ for all budgets, the overshoot term vanishes and $\mathrm{D_{MIB}} = 1 - \mathrm{AUC_{mag}}$. Negative values below 0 do not by themselves break the identity; only overshoot above 1 contributes the correction term. □

A.2 Isotonic calibration invariance

Let $u_{e_1}, \ldots, u_{e_n}$ be utilities sorted in descending order with fixed tie-breaking. If ϕ is strictly increasing, then

$$u_{e_i} > u_{e_j} \iff \phi(u_{e_i}) > \phi(u_{e_j}),$$

and equality is preserved as well. Therefore the total order induced by the utilities is unchanged under ϕ. A deterministic top-k selector with fixed tie-breaking depends only on that order, so its selected set is identical before and after transformation for every k. □

A.3 Zero interaction gap under exact additivity

Assume

$$f_m(C) = \frac{\sum_{e \in C} w_e^m}{\sum_{e \in \mathcal{E}} w_e^m}$$

for a non-negative edge utility system w^m. Let C_k be the set of the k highest-weight edges. By construction, C_k maximizes the numerator among all size-k circuits. Hence the additive top-k proxy and the realized faithfulness coincide exactly at budget k. Since this holds for every budget, the interaction gap is zero. Monotonicity follows because adding one more edge with non-negative weight cannot decrease the numerator. □

B Additional Results

Figure 6: Primary frontier view across representative cells.

Cell	Method	Pearson$_{\text{signed}}$	Pearson$_{\lvert.\rvert}$	Sign agr.	Recall@1%	Recall@5%
GPT-2 + IOI	EAP	0.814	0.817	0.969	0.870	0.890
GPT-2 + IOI	EAP-IG	0.931	0.932	0.698	0.755	0.644
GPT-2 + IOI	EAP-QK	0.590	0.591	0.940	0.650	0.658
GPT-2 + IOI	EAP-MLP	0.817	0.817	0.969	0.889	0.907
GPT-2 + IOI	EAP*	0.590	0.589	0.940	0.656	0.661
InterpBench + IOI	EAP	0.557	0.566	0.887	0.333	0.593
InterpBench + IOI	EAP-IG	0.644	0.664	0.685	0.167	0.704
InterpBench + IOI	EAP-QK	0.251	0.221	0.745	0.167	0.038
InterpBench + IOI	EAP-MLP	0.614	0.623	0.889	0.500	0.593
InterpBench + IOI	EAP*	0.273	0.245	0.746	0.167	0.077

Table 4: Score fidelity to exact edge effects on tractable cells. Higher exact-fidelity statistics do not reliably imply better downstream circuits.

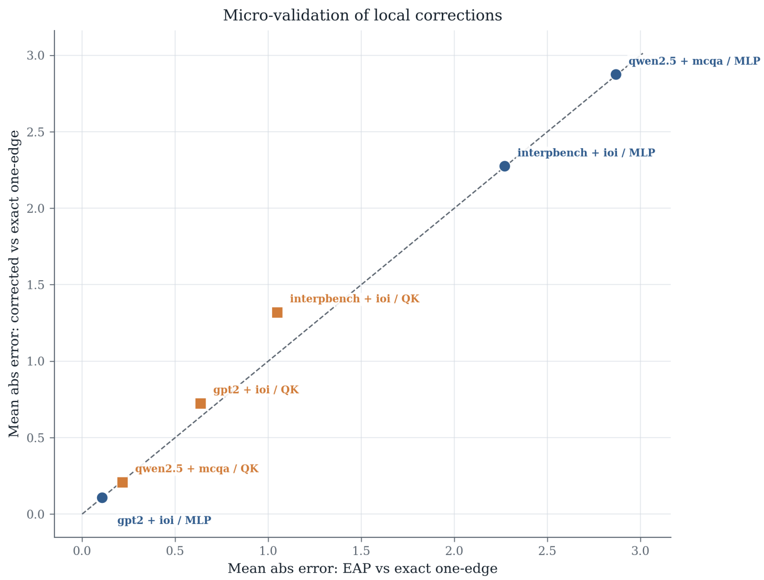

Figure 7: Micro-validation scatter for local corrections on sampled edges.

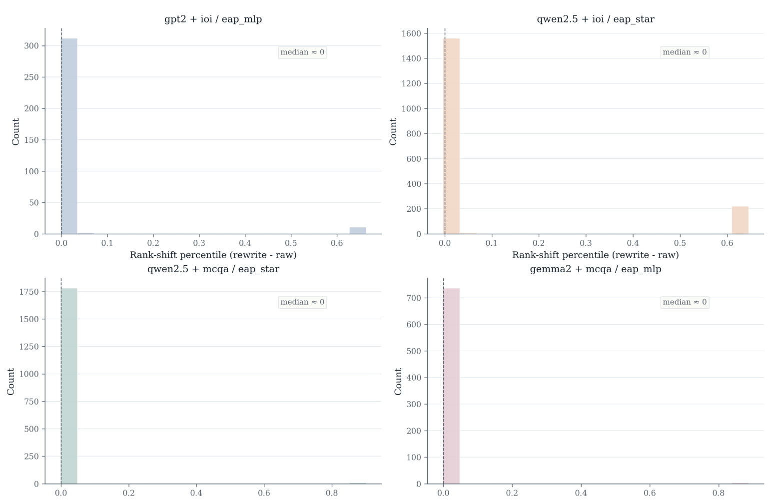

Figure 8: Rank-distortion histograms induced by calibration.

Figure 9: Selector-at-scale behavior across representative scorer-cell pairs.

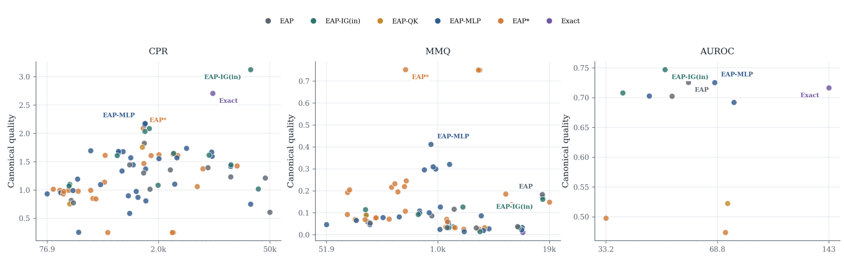

Figure 10: Global compute frontier across reporting metrics.

C Companion artifact

The companion artifact follows the paper's score-to-circuit framing. It contains the experimental stack used in the paper, exact-tractable probes, selector audits, compute records, theorem witnesses, and mechanized checks.

D Limitations and broader impacts

Limitations. The study centers on edge-level circuit localization under the intervention protocol and metric suite implemented by MIB. Exact-score analyses are limited to cells where exact one-edge patching is computationally tractable. Those exact references provide causal edge-level anchors; circuit-level conclusions still depend on the full score-to-circuit stack. The broader conclusions are strongest for the supported model–task matrix studied here. We report point estimates for most circuit metrics, with bootstrap statistics used primarily for calibration and selector design. Public-test transfer is reported for a selected set of frozen stacks after development, rather than as an exhaustive second campaign.

Broader impacts and release considerations. This work contributes methodological infrastructure for evaluating mechanistic localization. Better score-to-circuit evaluation can improve the reliability of interpretability claims, reduce overstatement from score-only analyses, and make comparisons between localization pipelines more scientifically meaningful. The same evaluation tools may lower the cost of identifying behaviorally important subgraphs in capable models, which could support targeted steering or intervention.